

DAR AL HAI GENERAL TRADING • HEAVY EQUIPMENT OPERATIONS

Parts Inquiry & SAP Business One PO API Integration

Operational Context, Komatsu PDX Allocation Rules, and Proposed SAP B1 Service Layer Integration

TARGET AUDIENCE	PREPARED BY	CORE SYSTEMS	DOCUMENT VERSION
ABS (SAP B1 Technical Team)	Motasem Ghanem (EQP Ops)	EQP Portal • Komatsu PDX • SAP B1	v1.0 (Integration RFC & Spec)

1. Executive Summary & Business Objective

Dar Al Hai is the authorized Komatsu equipment operator and service organization managing a major fleet of excavators, bulldozers, and heavy earthmoving machinery across infrastructure sites in Kuwait. To maintain operational readiness and rapidly restore breakdown machines, the Maintenance & Spare Parts department orders genuine replacement components directly from **Komatsu Middle East (KME / PDX Portal)**.

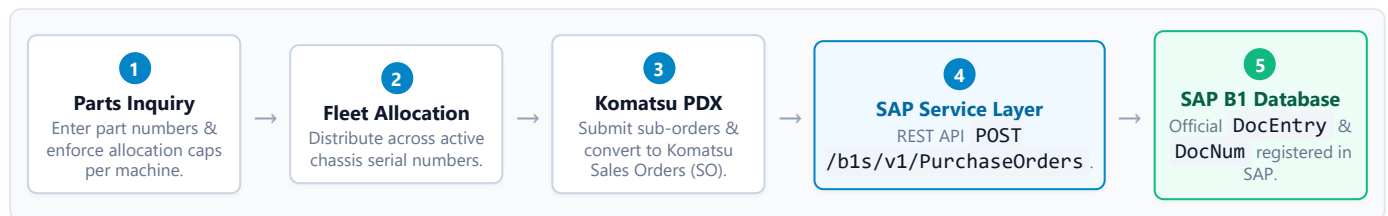
Once emergency orders are processed on the manufacturer portal, Komatsu confirms the stock and issues official **Komatsu Sales Orders (SOs)**. Each Komatsu Sales Order creates a direct commercial commitment billed by Vendor **V000006 (Komatsu Middle East)** in US Dollars (USD).

The Objective for the ABS Team:

We need to enable the EQP system to automatically create corresponding **Purchase Orders (POs)** in **SAP Business One** for Vendor **V000006** via the **SAP B1 Service Layer (REST API)**. This replaces our current temporary desktop RPA bridge, prevents manual entry delays, and ensures 100% accounting alignment between Komatsu billing and SAP financial ledgers.

2. End-to-End System Architecture

The target solution connects the field maintenance workflow directly to SAP B1 via automated API requests:



3. The Business Need: Why Orders Must Be Split

From an ERP perspective, purchasing 22 hydraulic hoses would normally be consolidated into a single PO line. However, the **Komatsu Middle East PDX Portal** enforces strict Emergency Order (EO) policies:

- **Mandatory Machine Chassis Binding:** Every emergency breakdown line item must be validated against a genuine Komatsu Model (e.g. **PC500LC-10R**) and registered Kuwait Chassis Serial Number (e.g. **100433**).
- **Anti-Hoarding Line-Item Quotas:** To prevent distributors from using breakdown channels to accumulate stock, Komatsu strictly caps the quantity per item per machine (e.g. **maximum 1 or 2 hoses per machine**).
- **Rejection of Bulk Orders:** Submitting a single order of 22 hoses against one excavator is **rejected by Komatsu portal validation**.
- **Sequential Reference Numbering:** Each dispatched sub-order must have a unique sequential distributor reference (e.g. **R229/2026 through R238/2026**).

The Operational Result:

To legitimately procure 22 hoses during a major emergency overhaul, the EQP system splits the request across **10 compatible machines**, resulting in **10 Komatsu Quotations**, which convert into **10 Komatsu Sales Orders**, and requires **10 corresponding Purchase Orders in SAP Business One**.

4. Step-by-Step Functional Workflow & Screenshots

Step 1: Requested Parts & Item Sub-Order Caps

The maintenance engineer enters required part numbers. The EQP engine verifies them against the genuine Komatsu catalogue, fetches official USD unit prices, and establishes the **Max / Sub-Order** constraint.

1 Requested Parts & Item Types
Add multiple item numbers. The automated dispatcher bundles line items together into unified quotations up to the sub-order max.

Import / Paste Verify Parts + Add Part Item

PART NUMBER	DESCRIPTION	REQUESTED QTY	MAX / SUB-ORDER	UNIT PRICE (\$)	LINE TOTAL (\$)
2A8-62-12230 ✓ Verified Genuine (PC500LC-10)	HOSE	12	2	66.310	\$795.72
2A8-62-11751 ✓ Verified Genuine (PC500LC-10)	HOSE	10	1	70.230	\$702.30
Total Summary: 2 Item Types		22 EA		Est. Total:	\$1498.020

Clear

Figure 1: Part Entry and Sub-Order Caps (Part 2A8-62-12230 capped at 2 EA/sub-order; Part 2A8-62-11751 capped at 1 EA/sub-order; Total: 22 EA, \$1,498.020).

Step 2 & 3: Fleet Machine Pool Allocation & Sequence Parameters

The dispatcher selects the customer fleet account (*Laala Al-Kuwait Real Estate Co.*), filters for model compatibility (*PC500LC-10R - 100% Match*), and activates a pool of genuine machines (20 serial numbers). The starting distributor reference is set (e.g., R229/2026) with breakdown remarks.

2 Target Fleet & Multi-SNs Allocation
Select target equipment serial numbers to distribute inquiry quotations

Customer Account: Laala Al-Kuwait Real Estate Machine Type: All Types (84) Model (Compatibility): ★ PC500LC-10R (100% Mat)

Serial Numbers in Pool: 20 ★ Auto-Select Compatible Select All (20) Clear Custom SNs Input

100461 (PC500LC-10R) Full Match	100463 (PC500LC-10R) Full Match
100466 (PC500LC-10R) Full Match	100468 (PC500LC-10R) Full Match
100477 (PC500LC-10R) Full Match	100478 (PC500LC-10R) Full Match
100480 (PC500LC-10R) Full Match	100504 (PC500LC-10R) Full Match
100510 (PC500LC-10R) Full Match	100520 (PC500LC-10R) Full Match

3 Order Sequence & PDX Settings
Starting order sequence, comments, and submission mode

Starting DB Order No: R229/2026

Sequential reference number

Order Comments / PO Reference: Urgent Breakdown

☐ Simulation Mode (Dry Run)
Calculate unified quotations and sub-orders without dispatching live records to Komatsu PDX.

Generated Sub-Orders: 10 Quotation(s)
Target Machine Pool: 20 SNs allocated
Est. Total Order Value: \$1498.020

Figure 2: Active Machine Pool Selection (20 SNs allocated) and Sequential Reference Generation.

Step 4: Planned Unified Quotations & Live Dispatch Queue

The dispatcher algorithm groups requested parts into compliant sub-orders, maximizing lines per order while respecting the individual quantity caps. Here, 22 units across 2 part numbers are allocated across 10 machines into **10 sub-orders** (R229/2026 to R238/2026) ready for live dispatch to Komatsu PDX.

4 Planned Unified Quotations & Live Dispatch Queue 10 sub-orders generated across 2 requested parts (22 units) Export CSV Export SAP Excel Dispatch All (10)								
#	DB ORDER NO	TARGET ASSET	BUNDLED PARTS REQUESTED	TOTAL QTY	EST. VALUE (\$)	QUOTATION #	STATUS	ACTION
1	R229/2026	100433 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	—	Ready	Inspect
2	R230/2026	100434 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	—	Ready	Inspect
3	R231/2026	100439 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	—	Ready	Inspect
4	R232/2026	100444 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	—	Ready	Inspect
5	R233/2026	100447 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	—	Ready	Inspect
6	R234/2026	100448 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	—	Ready	Inspect
7	R235/2026	100449 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA	1	\$70.230	—	Ready	Inspect
8	R236/2026	100454 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA	1	\$70.230	—	Ready	Inspect
9	R237/2026	100456 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA	1	\$70.230	—	Ready	Inspect
10	R238/2026	100459 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA	1	\$70.230	—	Ready	Inspect
View Technical Dispatch Audit Logs (4 entries)								

Figure 3: Unified Quotations Dispatch Queue displaying DB order numbers, target asset serial numbers, and line items.

Step 5: Komatsu Quotation to Sales Order (SO) Conversion & SAP Trigger

Once submitted to Komatsu PDX, the portal returns official Komatsu Quotation Numbers (e.g., 0000282871). The EQP system batch-confirms these quotations and converts them into official **Komatsu Sales Orders (SO)** (e.g., SO #0000278046 , SO #0000278047 ...).

Komatsu Quotation to Sales Order (SO) Converter

Batch confirm in-process quotations and copy them directly to Komatsu Sales Orders

Refresh

1-Click Convert (0)

Create SAP PO (0)

Export SAP Excel (0)

Search quotation #, customer, order #...

All (25)

Copied to SO Only (25)

Selected: 0 of 25

☐	QUOTATION #	DB ORDER NO	CUSTOMER	PERSON IN CHARGE	AMOUNT (USD)	STATUS	ACTIONS
☐	0000282871	R228/2026	DAR AL HAI	motasemgha	240.72	Transferred to SO ✓ SO #0000278046	Create SAP PO Unmark
☐	0000282870	R227/2026	DAR AL HAI	motasemgha	419.26	Transferred to SO ✓ SO #0000278047	Create SAP PO Unmark
☐	0000282869	R226/2026	DAR AL HAI	motasemgha	419.26	Transferred to SO ✓ SO #0000278048	Create SAP PO Unmark
☐	0000282863	225/2026	DAR AL HAI	motasemgha	1,383.84	Transferred to SO ✓ SO #0000278044	Create SAP PO Unmark
☐	0000282560	R224/2026	DAR AL HAI	mrami	549.90	Transferred to SO ✓ SO #0000277989	Create SAP PO Unmark
☐	0000282513	R223/2026	DAR AL HAI	mrami	116.00	Transferred to SO ✓ SO #0000277690	Create SAP PO Unmark
☐	0000282365	R222/2026	DAR AL HAI	mrami	142.20	Transferred to SO ✓ SO #0000277538	Create SAP PO Unmark
☐	0000282364	R221/2026	DAR AL HAI	mrami	245.08	Transferred to SO ✓ SO #0000277539	Create SAP PO Unmark
☐	0000282363	R220/2026	DAR AL HAI	mrami	245.08	Transferred to SO ✓ SO #0000277540	Create SAP PO Unmark
☐	0000282362	R219/2026	DAR AL HAI	mrami	245.08	Transferred to SO ✓ SO #0000277541	Create SAP PO Unmark
☐	0000282361	R218/2026	DAR AL HAI	mrami	245.08	Transferred to SO ✓ SO #0000277542	Create SAP PO Unmark

Figure 4: Komatsu Quotation to Sales Order Converter showing official Komatsu SO numbers and the target "Create SAP PO" action.

The Exact Integration Trigger:

At this stage, the Komatsu Sales Order is officially confirmed. Clicking **"Create SAP PO"** should call the SAP Service Layer API to instantly create the approved Purchase Order in SAP B1 for Vendor **V000006**, returning the SAP **DocNum** directly into this screen.

5. Technical API Specification for the ABS / SAP Team

To implement direct creation of Purchase Orders in SAP Business One, we propose integrating with the **SAP Business One Service Layer** (OData v4 REST API).

A. Authentication & Session Management

- **Endpoint:** POST `https://<sap-server>:50000/b1s/v1/Login`
- **Payload:** { "CompanyDB": "DAR_AL_HAI", "UserName": "API_USER", "Password": "..."} }
- **Session Handling:** SAP returns HTTP Session Cookies (B1SESSION and ROUTEID) which our backend preserves and attaches to subsequent requests.

B. Header Field Mapping (Table: OPOR / Entity: PurchaseOrders)

SAP FIELD	SAP DESCRIPTION	SOURCE FROM EQP SYSTEM	EXAMPLE VALUE
CardCode	Vendor Code	Komatsu Middle East Vendor Code	V000006
DocDate	Posting Date	Order Date (YYYY-MM-DD)	2026-10-05
DocDueDate	Delivery Due Date	Required Delivery Date	2026-10-12
DocCurrency	Document Currency	Komatsu Billed Currency	USD
NumAtCard	Vendor Ref / BP Reference	Komatsu SO Number + DB Order Ref	SO #0000278046 / R228/2026
Comments	Remarks	Breakdown Machine Model & Serial	Komatsu E0: PC500LC-10R (SN: 100433)
U_KomatsuSO	UDF: Komatsu SO No	Official Komatsu Sales Order Number	0000278046
U_MachineSerial	UDF: Machine Serial	Chassis Serial Number	100433
U_MachineModel	UDF: Machine Model	Equipment Model Code	PC500LC-10R

C. Line Item Field Mapping (Table: POR1 / Entity: DocumentLines)

SAP FIELD	SAP DESCRIPTION	SOURCE FROM EQP SYSTEM	EXAMPLE VALUE
ItemCode	Item Number	Genuine Komatsu Part Number	2A8-62-12230
ItemDescription	Item Description	Part Description from Komatsu Master	HOSE
Quantity	Quantity	Sub-order allocated quantity	2.00
UnitPrice	Unit Price (USD)	Komatsu Unit Price in USD	66.31
WarehouseCode	Warehouse	Receiving Warehouse	003 (or default per ABS)
TaxCode	Tax Code	Import Zero/Exempt	P0 (or default per ABS)

D. Concrete JSON Request Payload (`POST /b1s/v1/PurchaseOrders`)

Below is the exact JSON structure our backend will send to the Service Layer for each confirmed sub-order:

```
{
  "CardCode": "V000006",
  "DocDate": "2026-10-05",
  "DocDueDate": "2026-10-12",
  "DocCurrency": "USD",
  "NumAtCard": "SO #0000278046 / R228/2026",
  "Comments": "Komatsu Emergency Order | Asset: PC500LC-10R (SN: 100433) | Customer: Laala Al-Kuwait | Quote: 0000282871",
  "U_KomatsuSO": "0000278046",
  "U_KomatsuQuote": "0000282871",
  "U_MachineSerial": "100433",
  "U_MachineModel": "PC500LC-10R",
  "DocumentLines": [
    {
      "ItemCode": "2A8-62-12230",
      "ItemDescription": "HOSE",
      "Quantity": 2.0,
      "UnitPrice": 66.31,
      "WarehouseCode": "003",
      "TaxCode": "P0"
    },
    {
      "ItemCode": "2A8-62-11751",
      "ItemDescription": "HOSE",
      "Quantity": 1.0,
      "UnitPrice": 70.23,
      "WarehouseCode": "003",
      "TaxCode": "P0"
    }
  ]
}
```

E. Expected SAP Response & Storage

Upon successful creation, SAP B1 returns HTTP 201 Created with the document identifiers:

```
{
  "DocEntry": 14205,
  "DocNum": 20260481,
  "DocDate": "2026-10-05",
  "CardCode": "V000006",
  "DocTotal": 202.85,
  "DocCurrency": "USD"
}
```

The EQP database saves `DocNum: 20260481` and `DocEntry: 14205` alongside the Komatsu quotation record, giving the maintenance and finance teams instant cross-traceability.

6. Key Questions & Alignment Needed from ABS

1. Service Layer Endpoint URL & Connectivity:

Please provide the official Service Layer URL (e.g., `https://daralhai.b1pro.com:50000/b1s/v1/`) and confirm firewall access for the EQP backend server.

2. Dedicated Technical API User:

We request a dedicated API service account (e.g., `B1_API_EQP`) with authorizations limited to `PurchaseOrders` (Create/Read) to avoid relying on interactive user credentials.

3. Handling New Part Numbers (Item Master `0ITM`):

With over 500,000 Komatsu catalogue items, occasionally an emergency breakdown part does not yet exist in SAP B1. What is ABS's preferred procedure?

- *Option 1 (Recommended):* Allow our API to auto-create missing parts via `POST /b1s/v1/Items` using Komatsu catalogue metadata prior to PO creation.
- *Option 2:* Use a standard non-inventory item master with explicit line-item descriptions.

4. Document Status: Active PO vs. Draft:

Should the API create approved open POs (`PurchaseOrders`) directly, or create Drafts (`PurchaseOrderDrafts`) for accounting verification during the pilot phase?

5. Existing UDFs:

Please confirm if user-defined fields already exist on `OPOR` for Komatsu SO numbers or Machine Serials that we should populate.

Next Steps:

Once ABS provides the Service Layer connection parameters and confirms the Item Master policy, the EQP engineering team will execute test transactions in the SAP sandbox environment within 24 hours.